

Project/Research Paper: Computational Intelligence in a TOPIC

(A) Single or Team Project/Paper

Number of members: 1 – 3 people per group.

Note: If your team must have 4 members, a higher quality of the outcome/documentation, etc., is expected in the paper.

(B) Topic:

- Problem of your interest in applying the Soft Computing methods.
- If you already have a research topic and a concrete idea to apply SC to your topic, it's encouraged to work on your current topic.

(C) Potential Topics for Review/Application of {Neuro, Fuzzy, Probabilistic computing}: Open to any application area

Theory:

Enhancement of the existing NN/Fuzzy/Probabilistic model/algorithm/architecture, or the development of a new model/algorithm/architecture and its experimentation with the data set. e.g.) developing an improved fuzzy C-means algorithm in data clustering and its experiment.

Application:

- SC in Data Science: Data Analysis, Visualization, Deep learning, Machine learning, etc.
- SC in [Cyber Security & Cryptology](#)
- SC in (Web) Information Retrieval
- SC in Transfer Learning
- SC in Pattern Recognition – Image Processing
- SC in Bioinformatics, Biomedical Science
- SC in Medicine and Health Science
- SC in Control Engineering, Power System, Renewable Energy, etc.
- SC in Decision Analysis, Decision Making, Design, etc.
- SC in Operations Research, Financial Engineering, etc.
- SC in any Optimization problem
- Application of Fuzzy relations for reasoning, etc., in a certain problem domain
- SC applicable to the topic of your research
- Any others ++

(D) Review and Research Papers

(1) Proposal of a project/research.

- Refer to (A).
- The topic of the proposal may be changeable while you work on the 1st Review paper, but such a change is the final.

(2) A review paper of the related work for your topic by each computing method or its *hybrid* method.

- Neuro-computing, Fuzzy-computing, Probabilistic computing applied to your topic.
Requirement: Reviews must include studies of **at least two computing paradigms** in a balanced manner, e.g.) neuro-computing (50%) + fuzzy-computing (50%) or neuro-computing (40%) + fuzzy-computing (40%) + probabilistic-computing (20%).
- Hybrid computing: any *hybrid* method from {neuro, fuzzy, probabilistic, ML, evolutionary} where ML is any machine learning algorithm.

(i). Theoretical study.

(ii). Application of the (D.2.a, D.2.b, and/or D.2.c) methods in your topic.

(3) Final Paper.

Goal: Design (+ Implementation) an SC model for your topic or develop a (hybrid) SC model with the modified/new algorithm/architecture/etc. for performance improvement.

- a) **Highly Recommended: Fuzzy Computing** is highly recommended for your (hybrid) model.
- b) Theoretical study: The **Enhancement** of the existing algorithm/architecture/etc..
The *design* of a new algorithm/architecture/etc., and its *experiment* using the public data set, or their evaluation of a problem domain of your interest.
- c) Application: The design, implementation, and evaluation of your SC model for your topic.
- d) Review of literature **in-depth** on (hybrid) SC algorithms applied to your topic– an extension of (D.2) in-depth.
 - Not a combination of your review papers 1 & 2, but a review **in-depth** whose contents are > 60% newer than 1st & 2nd papers.

Thus, the review in (D.2) must be relevant/related, extendable, and applicable to (D.3).

(E) Format of the Papers

- Use the provided IEEE *conference* template and comply with it.
 - Because the initial goal is its submission to an IEEE conference based on the quality of your paper, use the ‘conference’ template. The IEEE transaction template may be used only if your paper is further extended for the journal.
- General structure of the review paper 1 & 2:

a) Introduction

The goal of the paper is to review the related works in C.(2).

Thus, do NOT spend more than a half page to describe any general or specific EA/SI algorithm.

b) Main part: The review of the related works in-depth in C.(2).

The review of related works in CSci 554 is not simply listing the related works to each independent problem, but requires a degree of depth to a topic consistently.

For example, in the review of the article [2],

What was the goal, and what was done in [2]?

Describe the computing model in [2]. – the overall scheme of the model.

What algorithm was applied? If a variant of the algorithm was applied and the outcome improved, describe that variant of the algorithm with the proper formulas, specifying the difference from the original algorithm.

How was the outcome improved? What needs to be further improved?

Any other important issue/fact, etc.

c) Conclusion

d) Reference

e) Number of pages:

Proposal: 1 – 2 page(s)

One review paper: 6 - 8 pages with full references.

Final paper: 6 – 8 pages with full references.

f) Figures/Tables:

In the review papers (2) and the final paper (3), the figures and tables must NOT exceed 3/4 page, in each paper.

If you need to include more figures and tables, add them as an appendix.

(F) Proposal & Final Paper:

- Single or Team project/paper of 1 – 3 members.
- Review Paper-1 & Paper-2: independent single work for each member of a team.
- Proposal & Final Paper: single or cooperative teamwork

(G) Formal Presentation:

- 20 – 25 minutes per group or a person
- Record your presentation with PPT Slide – See the instructions in BB.

(H) Deadline:

- Team Selection & Proposal: September 15th (Mon.)
 - A single submission per a team
- ~~Review Paper 1 with Neuro Computing: October 13th (Mon.)~~

- Review Paper & Presentation
- Final Paper & Presentation:

November 10th (Mon..)

December 8th (Mon.)

(I) Submission

1. Requirement:

- a) Paper: MS-Word or .pdf file.

The file name: **Proposal-ABC**, **Paper1-A**, ~~**Paper2-A**~~, **Final-ABC**, where **A, B, and C** are the **last names of members**.

- b) PPT slide of the presentation:

file name: Slide**1-A**, ~~Slide**2-A**~~, **Final-ABC.pptx**

- c) .mp4 file of a recording of the presentation.

File name: recording**1-A**, **Final-ABC.mp4**, etc.

- d) A compressed file of the entire reference papers: reference**1-A.zip**, etc.

OR In a separate file, include the titles of papers, doi: with the link: reference**1-A.docx**

e.g.) A Comparative Study of Different Soft Computing Techniques for Hybrid Renewable Energy Systems, **DOI:** [10.1109/ICOEI51242.2021.9453070](https://doi.org/10.1109/ICOEI51242.2021.9453070)

- e) In naming any of the files (a-d), your **last name(s)** (**A, B, C**) and the **paper number** (**1, 2, or Final**) must be included.

2. Submission Instruction:

- a) Submit everything in one .zip file.
- b) Prepare a folder, named: Paper1-A, ~~Paper2-A~~, Final-A, etc. with the same naming convention.
- c) Locate I.1.(a - d) to the folder
- d) **Compress** 2.c) to a .zip file – e.g.) Final-KimSmithMeyer.zip.
- e) Submit d) to the corresponding section in BB.

Or

Locate d) in One Drive of your UND account to be **shareable** and submit the link to it.

(J) List of Journals: check the BB.